

LEWWANDI VITHANAGE

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Objective

Software Engineer with a strong focus on AI/ML, experienced in building end-to-end pipelines with expertise in machine learning, deep learning, computer vision and GenAI –from model training to production deployment. Driven to contribute to innovative, real-world AI projects that create meaningful impact.

Experience

Zindhu Business Solutions (Pvt) Ltd

Apr 2025 – Present

Associate Software Engineer

- Designed and automated scalable backend infrastructure using Docker, Kubernetes, Helm, and Terraform; leveraged GitHub Actions to enable consistent CI/CD workflows.
- Developed an end-to-end real-time vehicle tracking system for an e-commerce platform, implementing React.js frontend, Node.js backend, and RESTful APIs for live GPS tracking and location updates.

Zindhu Business Solutions (Pvt) Ltd

Oct 2024 – Mar 2025

Intern Trainee

- Developed and tested RESTful APIs for backend services using Node.js, and contributed to the full-stack development of a responsive e-commerce platform with React.js.
- Configured and managed AWS cloud infrastructure including S3 buckets, IAM roles, and EKS deployments, gaining hands-on experience with cloud security and container orchestration.

Education

General Sir John Kotelawala Defense University

Jan 2022 – Dec 2025

BSc (Hons) in Electronics and Telecommunication Engineering

Ratmalana, Sri Lanka

Sirimavo Bandaranaike Vidyalaya

2012 – 2020

GCE Advanced Level - Physical Science Stream

Colombo 7, Sri Lanka

Projects

Production-grade Retrieval Augmented Generation Pipeline for Document QA System

Ongoing

- **Tools and Tech:** Python , LangChain , Pinecone , MLflow , RAGAS , Docker , Streamlit, OpenAI API
- **Motivation:** Engineered an end-to-end RAG pipeline using Pinecone and LangChain to enable natural language querying across documents with domain aware chunking,MMR-based retrieval.
- Implemented a production MLOps workflow with custom MLflow pyfunc wrapper to register,version and serve the LangChain RAG chain from the model registry by enabling zero downtime.
- Containerized full service stack(Streamlit UI,MLflow model server,MLflow tracking server) using Docker compose with health check gated startup sequencing.

Multi-Person Whole Body Human Mesh Reconstruction with Texture Mapping

Dec 2025

- **Tools and Tech:** PyTorch, PyTorch3D, SMPL-X, Stable Diffusion(SDXL), Metaverse
- **Motivation:** Generating textured human meshes from a monocular image for immersive VR gaming.
- Developing a pipeline using Global Z-Buffer depth-testing algorithm to eliminate "texture bleeding" in dense scenes.
- Fine-tuned an SDXL inpainting model to synthesize realistic textures for occluded regions using custom masks and executed morphological texture padding with backface culling for seamless UV surface rendering.

Multi Tenant SaaS Platform for Ayurvedic Wellness Resorts

Feb 2026

- **Tools and Tech:** React.js, Node.js , MongoDB(replica sets) , AWS , Docker , Kubernetes , Helm , Terraform , Grafana , Github Actions

- **Motivation:** Built a cloud-native ,cluster per tenant Saas platform consisting 26+ deployable microservices orchestrated on AWS EKS with mongoDB replica sets.
- All served through a centralized API gateway with rate limiting and Grafana monitoring dashboard for checking APIs.
- Developed a fully automated Provisioning API that uses Terraform and Helm to spin up a dedicated EKS cluster, configure IRSA roles, deploy all services, and initialize a 3-node MongoDB replica set for each new resort tenant from a single API call, with orphan-recovery logic for failed provisioning runs.

Sentiment Analysis Web Application

May 2025

- **Tools and Tech:** Python, Flask, NLTK, Scikit-learn, TF-IDF, Logistic Regression
- **Motivation:** To build a robust multiclass sentiment classifier enabling data-driven insights from text data.
- Optimized Logistic Regression, Linear SVM and Random Forest using GridSearchCV with 5-fold stratified cross-validation, tuning hyperparameters including L1/L2 regularization
- Deployed Flask REST API with real-time prediction endpoint to deliver sentiment classification on user-generated text

Heart Disease Prediction Web Application

Jan 2025

- **Tools and Tech:** Python, Pandas, Flask, JSON Web Tokens
- **Motivation:** To provide early risk assessment support for heart diseases using accessible clinical data.
- Developed a full-stack web application to predict heart disease risk by analyzing 11 clinical parameters.
- Applied secure authentication using JSON Web Tokens for user login and registration.

Voice Assisted AI Chatbot on Raspberry Pi: Embedded AI System

Dec 2023

- **Tools and Tech:** Python, Whisper, Google Gemini API, pytttsx3, Raspberry Pi
- **Motivation:** To explore deploying conversational AI systems on resource-constrained embedded hardware.
- Enhanced a voice-controlled chatbot integrating Whisper-based speech recognition and Gemini LLM on Raspberry Pi.

Skills

Languages & Tools: Python, MERN Stack, JavaScript, C/C++, Bash, Git, Linux

AI & Computer Vision: PyTorch, PyTorch3D, TensorFlow, Scikit-learn, OpenCV, SMPL-X, LoRA, Pinecone, MLflow, LangChain , MLOps

Cloud & DevOps: AWS, Docker, Kubernetes, Helm, Terraform,CI/CD (Github Actions)

Gen AI: Diffusion Models(SDXL),Texture Inpainting

Leadership Experience

Electronic, Robotics & Innovation Club-KDU	2023 – 2024
Marketing Director	
IEEE Communications Society-KDU	2023 – 2024
Program Team Member	
Toastmasters Club-KDU	2022 – 2023
Education Team Member	

Additional Information

Languages: English, Sinhala (Native)

References

Prof T.L. Weerawardhane	Pradeep Indrajith
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